

# KUMAR VARENYA

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## EDUCATION

### University of British Columbia

Vancouver, BC

*Master of Applied Science in Electrical and Computer Engineering*

2026 – 2028

*Bachelor of Applied Science in Computer Engineering & Honours Mathematics*

2021 – 2026

## EXPERIENCE

### Hootsuite

Toronto, ON

*Software Engineer*

April 2026 – Present

- Designing and building scalable, cloud-native backend microservices utilizing Scala, Golang, and Docker within an AWS (EKS, EC2) environment.
- Implementing event-driven streaming architectures with Apache Kafka to optimize high-throughput social media data processing.
- Collaborating within an agile team to enhance CI/CD pipelines, ensure code quality, and monitor system reliability using Prometheus and Grafana.

### UBC Investment Management Trust Inc.

Vancouver, BC

*Risk/Systems Engineer (UBC CS DMM: Operations Research & Financial Engineering)*

September 2025 – Present

- Developed quantitative optimization models for allocation across multi-asset \$2.336 billion endowment portfolio.
- Engineered risk analytics infrastructure integrating factor models, scenario simulations, and stress-testing.
- Collaborated with UBCiM to translate operational research outputs into actionable allocation and hedging strategies.

### Amazon

Vancouver, BC

*Software Development Engineer Intern (AWS)*

May 2024 – December 2024

- Built AI software package using Docker, Kubernetes, and AWS (CDK, Fargate, Bedrock), improving efficiency by 50%.
- Designed secure LLM prompt templates and PII detection, cutting errors by 70% with zero breaches.
- Implemented unit and integration tests (100% pass rate); contributed to key feature deployment for Prime Day.

### Population Data BC

Vancouver, BC

*Software Programmer*

September 2022 – December 2025

- Built Python and SQL ETL pipelines to boost data throughput and reduce latency for large epidemiological datasets.
- Developed containerized microservices with API gateways to enable secure, compliant data access.
- Refactored legacy code using design patterns and optimizations to improve scalability and maintainability.

### University of British Columbia, Department of Computer Science

Vancouver, BC

*Teaching Assistant (CPSC 110, CPSC 121)*

January 2023 – April 2024

- Led weekly tutorials for 2000+ students, improving understanding of functional programming and discrete math.
- Graded over 500 assignments and exams per week and provided constructive feedback to improve student performance.
- Collaborated with course instructors to update course materials, aligning content with current programming practices.

### Oil and Natural Gas Corporation

New Delhi, IN

*Summer Intern*

Summer 2022, Summer 2023

- Analyzed pipeline monitoring data to identify potential failures, reducing unplanned maintenance incidents by 20%.
- Supported the implementation of GIS mapping tools for resource allocation and logistics planning.
- Assisted in designing database systems to track over 1,000 equipment inspections and maintenance schedules.

## RESEARCH EXPERIENCE

### Carnegie Mellon University

Pittsburgh, PA

*Undergraduate Research Assistant*

May 2025 – July 2025

- Investigating compositional verification techniques for concurrent software systems to enhance modular analysis.
- Applied static analysis to identify data races and synchronization bugs in multithreaded C/C++ programs.

### UBC Pacific Laboratory of Artificial Intelligence

Vancouver, BC

*Data Engineering Research Assistant*

September 2023 – April 2024

- Collected, curated, and annotated Minecraft gameplay datasets to support behavioral and AI-driven analysis.
- Built preprocessing pipelines to structure spatial-temporal data for downstream machine learning tasks.

## TECHNICAL SKILLS

**Languages:** Java, C/C++, Python, R, SQL (PostgreSQL), HTML/CSS, MATLAB, Lua, ARM Assembly, Verilog

**Frameworks & Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Flask, JUnit, Swing, JavaFX

**Tools & Platforms:** Docker, Linux, Git, Jupyter, CMake, Bash & Shell, Valgrind, Arduino, Raspberry Pi, Altium Designer

**AI & Workflows:** Prompt Engineering, Agentic Workflows, Cursor, Claude Code